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Ruiyang Li
Department of Aerospace and Mechanical Engineering, University of Notre Dame, Notre Dame, IN 46556

Jiahang Zhou
Department of Aerospace and Mechanical Engineering, University of Notre Dame, Notre Dame, IN 46556

Jian-Xun Wang
Department of Aerospace and Mechanical Engineering, University of Notre Dame, Notre Dame, IN 46556; Center for Sustainable Energy at Notre Dame (ND Energy), University of Notre Dame, Notre Dame, IN 46556; Lucy Family Institute for Data & Society, University of Notre Dame, Notre Dame, IN 46556

Tengfei Luo[1]
Department of Aerospace and Mechanical Engineering, University of Notre Dame, Notre Dame, IN 46556; Center for Sustainable Energy at Notre Dame (ND Energy), University of Notre Dame, Notre Dame, IN 46556; Department of Chemical and Biomolecular Engineering, University of Notre Dame, Notre Dame, IN 46556
e-mail: tluo@nd.edu

Physics-Informed Bayesian Neural Networks for Solving Phonon Boltzmann Transport Equation in Forward and Inverse Problems With Sparse and Noisy Data

Nondiffusive phonon transport presents significant challenges in micro/nanoscale thermal characterization, compounded by the limitations of experimental-numerical techniques and the presence of measurement noise. Additionally, inverse modeling and uncertainty quantification (UQ) for submicron thermal transport remain under-explored. In this study, we introduce a physics-informed Bayesian deep learning framework designed to address phonon Boltzmann transport equation (BTE)-based forward and inverse problems leveraging limited and noisy data. Our approach combines Bayesian neural networks with a nonparametric variational inference method, formulating the BTE-constrained training in a Bayesian manner. This enables the estimation of the posterior distribution of neural network parameters and unknown equation parameters based on a likelihood function that incorporates uncertainties from both the measurement data and the BTE model. Through numerical experiments on various phonon transport scenarios, we demonstrate that our method can accurately reconstruct temperature and heat flux profiles, infer critical quantities of interest (e.g., Knudsen number), and provide robust uncertainty quantification, even when data is sparse and noisy. This framework enhances our capability to conduct nondiffusive thermal simulations and inverse modeling with quantified uncertainty, offering a powerful tool for advancing thermal transport research and optimization in micro/nanoscale devices. [DOI: 10.1115/1.4067163]

1 Introduction

Micro-electronics has become the core of modern electronic products, powering applications ranging from artificial intelligence computing to advanced industrial systems. High electron mobility transistors, for instance, are widely used in high-frequency and high-voltage switching applications [1]. Their miniaturization leads to significantly larger power densities, resulting in elevated operating temperatures [2], degrading performance and reliability, and posing a roadblock for their further advancements [3]. Accurately evaluating channel temperature and thermal properties is crucial for understanding the micro/nanoscale thermal transport in these devices and mitigating the self-heating effects. However, characterizing thermal transport in micro- and nanostructures is particularly challenging. When the characteristic dimensions, such as thin film thickness and fin width, approach or fall below the mean free path of heat carriers (electrons or phonons), thermal transport deviates from the purely diffusive behavior, rendering the conventional Fourier's law inadequate [4].

Various methods for simulating thermal transport at micro/nanoscale have emerged, including lattice dynamics [5], atomistic Green's function [6], molecular dynamics [7], and the phonon Boltzmann transport equation (BTE) [8]. Among these, the phonon BTE is notably effective for describing multiscale thermal transport from the ballistic to diffusive regimes. Given the computational impracticality of atomistic-level simulation techniques for device-level thermal analysis [9], mesoscopic approaches based on the phonon BTE offer a balanced solution between computational complexity and accuracy. The phonon BTE can be solved using several numerical methods, such as the Monte Carlo method [10],

[1]Corresponding author.

Contributed by the Heat Transfer Division of ASME for publication in the Journal of Heat and Mass Transfer. Manuscript received August 27, 2024; final manuscript received October 2, 2024; published online December 16, 2024. Assoc. Editor: Bolin Liao.

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implicit discrete ordinate method [11], and discrete unified gas kinetic scheme [12]. However, deterministic solvers often struggle with ill-posed problems, particularly those involving unknown parameters and boundary conditions. Additionally, simulation results alone may be insufficient or invalid due to the inherent uncertainties in device and material properties, complicating the estimation of prediction uncertainties. On the other hand, experimental techniques such as infrared microscopy [13], Raman thermometry [14], and thermoreflectance techniques [15,16] have been widely employed for thermal property characterization and thermal imaging. Each of these techniques, however, presents limitations related to accessibility, resolution, and measurement accuracy [17,18]. Ideally, thermal measurements can validate numerical simulations and be used for reversely identifying key thermal parameters of the system. Nonetheless, measurement data are often limited and inevitably contain noise, making it difficult to fully leverage the combined experimental–numerical investigation while providing reliable uncertainty quantification (UQ).

Recent advances in machine learning techniques have opened new avenues for developing efficient algorithms for thermal characterization from limited measurements. Deep neural networks (DNNs), for example, have been employed to learn thermal properties using data from measurements or calculations [19]. However, the effectiveness of purely data-driven approaches relies heavily on the amount and quality of training data, which can lead to poor performance with sparse or noisy datasets. As a robust alternative, physics-informed neural networks (PINNs) have been proposed to solve both forward and inverse problems for various partial differential equations [20]. By encoding underlying physical laws in the form of partial differential equations into the loss function, PINNs constrain the solution space and infer unknown terms from limited data. PINNs offer several advantages over traditional numerical solvers, such as providing closed-form solutions, being mesh-free (enabling on-demand solution computation post-training), and discovering new solutions quickly through parametric learning [21]. In the context of micro/nanoscale thermal transport, PINNs have been adopted for forward simulation by solving mode-resolved phonon BTE [22–24] and coupled electron-phonon BTE [25].

Although physics-informed deep learning shows great promise for addressing forward phonon BTE-based problems in thermal modeling with limited data, it faces challenges when dealing with measurement noise and model-form uncertainties since traditional PINNs are typically formulated in a deterministic manner [21]. Furthermore, existing PINN-based models have not been thoroughly validated for inverse nondiffusive heat conduction problems, where quantities of interest must be inferred from temperature observations or other thermal measurements. To fully realize the potential of PINNs, it is crucial to develop models that can effectively leverage sparse, noisy data while supporting UQ.

Recently, Bayesian learning approaches have been integrated into the PINN framework to enable UQ in both forward and inverse problems [26,27]. These enhanced models, known as physics-informed Bayesian neural networks (PI-BNNs), leverage the strengths of PINNs in creating parameterized surrogate models while incorporating scalable Bayesian inference for estimating posterior distributions. In this framework, the physics-constrained training is formulated in a Bayesian manner, where the posterior distribution of the trainable parameters is obtained based on a likelihood function accounting for uncertainties from both measurement noise and physical constraints. This integration enables UQ by providing posterior distribution estimates for the predictions. PI-BNNs have been successfully applied to fluid flow reconstruction governed by the Navier–Stokes equation [26], utilizing Stein variational gradient descent (SVGD) [28] for scalable Bayesian learning.

Building on recent advancements, we present a physics-informed deep learning framework specifically designed for forward simulation and unknown parameter inference in nondiffusive thermal modeling using sparse and noisy data. Our method addresses both the forward and inverse solutions of the phonon BTE while quantifying uncertainties by employing Bayesian neural networks as a parameterized surrogate model to approximate phonon energy distributions. To account for uncertainties, the PI-BNN framework initializes with prior distributions for both network parameters and any unknown equation parameters, which are updated to form posterior distributions based on the physics-informed likelihood through Bayesian learning. Specifically, the likelihood function is defined as a joint conditional probability distribution that incorporates both data uncertainty and BTE model-form uncertainty, each modeled as a Gaussian distribution. To enable scalable Bayesian inference, SVGD is utilized to estimate the posterior distributions of the surrogate model parameters and, if needed, the equation parameters. Several numerical benchmark problems are studied to demonstrate the effectiveness of PI-BNNs in solving both forward and inverse phonon transport problems. Our framework leverages the strengths of PINNs and variational Bayesian inference, providing a robust method for accurately modeling nondiffusive thermal transport with sparse and noisy data.

2 Methods

2.1 Phonon Boltzmann Transport Equation. Under the single-mode relaxation time approximation, the phonon BTE describing the transport of phonons can be written as [4]

$$\frac{\partial f}{\partial t} + \mathbf{v} \cdot \nabla f = -\frac{f - f^{\mathrm{eq}}}{\tau} + \delta \tag{1}$$

where $f(\mathbf{x}, \mathbf{k}, p, \mathbf{s}, t)$ is the phonon distribution function dependent on the spatial vector $\mathbf{x}$, wavevector $\mathbf{k}$ (or frequency ω), polarization p, solid angle $\mathbf{s}$, and time t. $\mathbf{v}$ is the group velocity and τ is the relaxation time. f^{eq} represents the phonon equilibrium distribution following the Bose-Einstein distribution. Scattering interactions other than the intrinsic phonon-phonon scattering are incorporated into the source term δ. As the primary heat carrier in semiconductors is phonon, the phonon BTE effectively describes the multiscale thermal transport process from ballistic to diffusive.

For convenience, by defining the phonon energy density distribution $e = \hbar\omega D(\omega, p)[f - f^{\mathrm{eq}}(T_{\mathrm{ref}})]$ and the associated equilibrium distribution function $e^{\mathrm{eq}} = \hbar\omega D(\omega, p)[f^{\mathrm{eq}} - f^{\mathrm{eq}}(T_{\mathrm{ref}})]$, where T_{ref} is the reference temperature and $D(\omega, p)$ is the phonon density of states, we rewrite Eq. (1) in the energy-based form as follows:

$$\frac{\partial e}{\partial t} + \mathbf{v} \cdot \nabla e = -\frac{e - e^{\mathrm{eq}}}{\tau} + \varphi \tag{2}$$

where φ is the volumetric heat generation term. Assuming that the temperature difference throughout the domain is much smaller than the reference temperature ($|T - T_{\mathrm{ref}}| \ll T_{\mathrm{ref}}$), we can approximate the equilibrium energy density as [29]

$$e^{\mathrm{eq}} \approx \hbar\omega D(\omega, p) \frac{\partial f^{\mathrm{eq}}}{\partial T}(T_{\mathrm{ref}})(T - T_{\mathrm{ref}}) = C(T - T_{\mathrm{ref}}) \tag{3}$$

Here, $T = T_{\mathrm{ref}} + e^{\mathrm{eq}}/C$ is the lattice temperature, and C is the mode specific heat at T_{ref}. Considering the energy conservation during phonon scattering, we have

$$\sum_p \int_0^{\omega_{\max,p}} \int_{4\pi} \frac{e - e^{\mathrm{eq}}}{\tau} d\Omega d\omega = 0 \tag{4}$$

where Ω and $\omega_{\max,p}$ are the solid angle space and maximum frequency for a given phonon branch p, respectively.

Combining Eqs. (3) and (4), we can obtain the lattice temperature from e. Moreover, the heat flux is calculated by

$$\mathbf{q} = \sum_p \int_0^{\omega_{\max,p}} \int_{4\pi} \mathbf{v} e d\Omega d\omega \tag{5}$$

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Commonly employed boundary conditions, including the isothermal boundary, the specularly/diffusely reflecting boundary, and the periodic boundary, are discussed in Appendix.

2.2 Physics-Informed Neural Networks. Standard PINNs approximate the unknown solution of the governing PDE using a DNN parameterized by $\boldsymbol{\theta}$ (e.g., weights and bias). Unlike purely data-driven machine learning algorithms, PINNs leverage physical laws to constrain the training process, thereby incorporating additional domain knowledge. This approach involves training the DNN on sparse data while imposing physical laws at a number of spatiotemporal collocation points where measurements are not available. Consequently, the trained DNN becomes a smooth function in the spatiotemporal domain, capable of reconstructing the solution with high resolution. Training PINNs is typically formulated as a deterministic optimization problem, where the loss function combines data mismatch with the residuals of the governing equations, such as the phonon BTE. By minimizing this physics-informed loss, the solution is expected to adhere to physical principles and match the available measurement data. This integration of data and physics ensures that the surrogate model provides accurate and physically consistent predictions, even with sparse observational data. It is worth noting that, in the extreme case, training PINNs can solely rely on the PDE residuals without the need of any labeled data.

Our previous studies have introduced several PINN designs for solving the phonon BTE under various conditions [22–25]. These designs utilize either two independent networks or a single network with two outputs to separately learn the equilibrium (e^{eq}) and nonequilibrium ($e^{\mathrm{neq}} = e - e^{\mathrm{eq}}$) components of the phonon energy distribution. The two outputs are then combined to form the total energy distribution e. To ensure energy conservation as described in Eq. (4), we leverage the flexibility of the DNN output. Specifically, we define the operation in Eq. (4) as

$$\langle e^{\mathrm{neq}} \rangle := \sum_p \int_0^{\omega_{\max,p}} \int_{4\pi} \frac{e^{\mathrm{neq}}}{\tau} d\Omega\, d\omega = 0 \tag{6}$$

with $\langle 1 \rangle = \sum_p \int_0^{\omega_{\max,p}} \int_{4\pi} \frac{1}{\tau} d\Omega\, d\omega$. Assuming that z is one of the two network outputs, we let $e^{\mathrm{neq}} = z - \langle z \rangle / \langle 1 \rangle$ such that it automatically satisfies Eq. (6). Furthermore, for linear transport equations like BTE that exhibit multiscale or diffusive behavior, an additional loss term with the asymptotic-preserving property is necessary [30]. This involves integrating Eq. (2) over solid angle space and wavevector space, yielding

$$\frac{\partial E}{\partial t} + \nabla \cdot \mathbf{q} = \Phi \tag{7}$$

where E and Φ represent the integration of phonon energy density and the heat generation term, respectively.

From Eqs. (2) and (7), we can derive the residual function $\mathcal{R}$ of the phonon BTE within the physical domain Γ given the equation parameters $\boldsymbol{\lambda}$ (e.g., characteristic length of the system)

$$\mathcal{R}(e;\lambda) := \begin{cases} \dfrac{\partial e}{\partial t} + \mathbf{v} \cdot \nabla e + \dfrac{e^{\mathrm{neq}}}{\tau} - \varphi = 0, \\ \dfrac{\partial E}{\partial t} + \nabla \cdot \mathbf{q} - \Phi = 0, \end{cases} \quad \mathbf{x}, \mathbf{k}, p, \mathbf{s}, t \in \Gamma \tag{8}$$

In the deterministic scenario with known $\boldsymbol{\lambda}$, proper initial and boundary conditions (I/BCs) are required to determine unique solutions

$$\begin{cases} \mathcal{I}(e;\boldsymbol{\lambda}) = 0, & t = 0, \text{ in } \Gamma \\ \mathcal{B}(e;\boldsymbol{\lambda}) = 0, & \text{on } \Gamma_b \end{cases} \tag{9}$$

where $\mathcal{I}$ is an initial condition operator, and $\mathcal{B}$ is a boundary condition operator acting on the domain boundary Γ_b. Four typical boundary conditions are reviewed in Appendix. In this work, soft I/BC enforcement is applied, incorporating I/BCs as penalty terms into the loss function. When measurement data (e.g., temperature at various locations) are available, the discrepancy between the DNN predictions and the measurements can be captured by a data-based loss component, $\|g(e) - g_d\|$, where g represents an observable function of the predicted phonon energy e, and g_d denotes the corresponding observations from thermal measurement.

Finally, the loss function for the phonon BTE can be written as follows:

$$\mathcal{L}(\boldsymbol{\theta}) = \xi_1 \|\mathcal{R}(e;\boldsymbol{\lambda})\|^2 + \xi_2 \|\mathcal{I}(e;\boldsymbol{\lambda})\|^2 + \xi_3 \|\mathcal{B}(e;\boldsymbol{\lambda})\|^2 + \xi_4 \|g(e) - g_d\|^2 \tag{10}$$

where $\|\cdot\|$ denotes the L_2 norm, and ξ_1, ξ_2, ξ_3, ξ_4 are the penalty coefficients that can be adjusted to optimize training. All the derivative terms in $\mathcal{R}$ are computed using automatic differentiation. The network parameters are typically optimized using the adaptive moment estimation (Adam) optimizer [31], a robust variant of the stochastic gradient descent algorithm.

2.3 Physics-Informed Bayesian Neural Networks. In our earlier work, PINNs demonstrated the capability to perform forward simulations based on the phonon BTE with minimal or no data. However, they did not account for uncertainties arising from imperfect physical models, unknown parameters, and noisy data. To better quantify these impacts on neural network outputs, a probabilistic formulation is necessary, wherein training is conducted in a Bayesian manner. This involves initializing the DNN with network parameters $\boldsymbol{\theta}$ that follow a specific prior distribution $p(\boldsymbol{\theta})$. In the forward problem setup, we construct the likelihood function $p(\mathcal{D}, \mathcal{R}|\boldsymbol{\theta})$, which is defined by the uncertainties in sparse data $\mathcal{D} = \{g_d\}$ and the physical constraints $\mathcal{R} = 0$. Using Bayes' rule, the posterior distribution $p(\boldsymbol{\theta}|\mathcal{D}, \mathcal{R})$ is obtained

$$p(\boldsymbol{\theta}|\mathcal{D}, \mathcal{R}) \propto p(\mathcal{D}, \mathcal{R}|\boldsymbol{\theta}) p(\boldsymbol{\theta}) \tag{11}$$

By sampling from this posterior, the trained DNNs can provide mean predictions along with estimated uncertainties. While the Bayesian method is not explicitly used for optimizing hyperparameters such as learning rates or network architectures, it plays a critical role in guiding the optimization of the model parameters within the PI-BNN framework, ultimately improving both the robustness of predictions and the quantification of uncertainties.

In many practical applications, computing the posterior distribution is intractable due to the high dimensionality of the parameter space. To address this, variational inference (VI) [32] is employed to approximate the target distribution with a proposed density function parameterized by a specific form of distributions. However, the performance of parametric VI is highly dependent on the predefined family of distributions, which introduces deterministic biases. In this work, we utilize a nonparametric VI method, SVGD [28], to approximate the posterior distribution with a set of particles $\{\boldsymbol{\theta}_j\}_{j=1}^n$. The core idea is to iteratively move the set of particles toward the posterior distribution using the gradient ϕ of the Kullback-Leibler (KL) divergence. The update rule for these particles involves the kernelized Stein operator to identify the steepest descent direction for reducing the KL divergence

$$\phi(\boldsymbol{\theta}) = \frac{1}{n} \sum_{j=1}^{n} [\mathrm{k}(\boldsymbol{\theta}_j^t, \boldsymbol{\theta}) \nabla_{\boldsymbol{\theta}_j^t} (\log p(\boldsymbol{\theta}_j^t) + \log p(\mathcal{D}, \mathcal{R}|\boldsymbol{\theta}_j^t)) + \nabla_{\boldsymbol{\theta}_j^t} k(\boldsymbol{\theta}_j^t, \boldsymbol{\theta})] \tag{12}$$

$$\boldsymbol{\theta}_j^{t+1} = \boldsymbol{\theta}_j^t + \epsilon_t \phi(\boldsymbol{\theta}_j^t) \tag{13}$$

where ϵ_t represents the step size at the t-th iteration, and $k(\cdot, \cdot)$ is a positive definite kernel function (e.g., radial basis function used in

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this work). Consequently, an ensemble of DNNs corresponding to parameter particles is trained using SVGD.

We assume that the model inadequacy in physical constraints (i.e., residuals of governing equations) and the measurement errors in sparse observation data both follow zero-mean Gaussian distributions, i.e., $p(\mathcal{R}|\boldsymbol{\theta}) \sim \mathcal{N}(0, \Sigma_{\mathcal{R}})$ and $p(\mathcal{D}|\boldsymbol{\theta}) \sim \mathcal{N}(0, \Sigma_{\mathcal{D}})$. Consequently, the joint log likelihood function can be expressed as

$$\log(\text{likelihood}) = \log p(\mathcal{D}|\boldsymbol{\theta}, \Sigma_{\mathcal{D}}) + \log p(\mathcal{R}|\boldsymbol{\theta}, \Sigma_{\mathcal{R}}) \tag{14}$$

where the covariance matrices are $\Sigma_{\mathcal{R}} = \text{diag}\left(\sigma_{\mathcal{R}}^2\right)_{n\times n}$ and $\Sigma_{\mathcal{D}} = \text{diag}\left(\sigma_{\mathcal{D}}^2\right)_{n\times n}$, respectively. $\Sigma_{\mathcal{R}}$ reflects the confidence in the physical model. Given that the phonon BTE generally provides an accurate description of phonon transport, a small variance ($\sigma_{\mathcal{R}}^2 = 1 \times 10^{-4}$) is specified in this work. On the other hand, $\Sigma_{\mathcal{D}}$ is treated as a trainable parameter, with the data variance $\sigma_{\mathcal{D}}^2$ modeled using an inverse Gamma distribution. The physics-informed SVGD algorithm is summarized in Algorithm 1.

Algorithm 1. PI-BNN based on Stein variational gradient descent for the phonon BTE

Input: A set of sparse, noisy thermal measurement data $\{g_d\}$.

Output: DNN parameters $\{\boldsymbol{\theta}_j\}_{j=1}^n$ (including equation parameters $\boldsymbol{\lambda}$ in inverse problems) and learnable data variance $\Sigma_{\mathcal{D}}$.

for $t = 0 : n_t$ **do**

1. Calculate the posterior:
   $\mathcal{L}(\boldsymbol{\theta}^t) = \log p\left(\mathcal{R}|\boldsymbol{\theta}^t, \Sigma_{\mathcal{R}}\right) + \log p\left(\mathcal{D}|\boldsymbol{\theta}^t, \Sigma_{\mathcal{D}}\right) + \log p(\Sigma_{\mathcal{D}}) + \log p(\boldsymbol{\theta}^t)$
2. Calculate the gradient $\nabla_{\boldsymbol{\theta}^t}$ and obtain $\phi(\boldsymbol{\theta}^t)$
3. Update $\boldsymbol{\theta}$ and $\Sigma_{\mathcal{D}}$ by stochastic gradient descent (e.g., Adam)

end for

After training, the network ensemble within the PI-BNN framework can provide a mean prediction of the phonon energy distribution, ultimately yielding the lattice temperature and other thermal properties. The corresponding posterior variance, which reflects reconstruction uncertainty, is computed as follows:

$$\text{Var}(g|\mathbf{x}, t, \mathcal{D}) \approx \sigma_{\mathcal{D}}^2 + \frac{1}{n}\sum_{i=1}^{n}\left(g - \mathbb{E}_{\boldsymbol{\theta}|\mathcal{D}}(g)\right)^2 \tag{15}$$

where g is an observable quantity derived from the phonon energy e. The total variance combines the uncertainty estimate from the posterior distribution of network parameters (represented by n particles) with the predicted data uncertainty due to measurement noise.

For inverse problems where certain physical parameters $\boldsymbol{\lambda}$ in the equation are unknown, we can formulate the PI-BNN model in a similar manner. In addition to the network parameters $\boldsymbol{\theta}$, we should also assign a prior distribution for the equation parameters $\boldsymbol{\lambda}$, which could be independent of $p(\boldsymbol{\theta})$. The likelihood remains the same as in Eq. (14), but it now includes an additional parameter vector $\boldsymbol{\lambda}$. Consequently, the posterior is calculated as

$$p(\boldsymbol{\theta}, \boldsymbol{\lambda}|\mathcal{D}, \mathcal{R}) \propto p(\mathcal{D}, \mathcal{R}|\boldsymbol{\theta}, \boldsymbol{\lambda})p(\boldsymbol{\theta})p(\boldsymbol{\lambda}) \tag{16}$$

Since forward and inverse problems are formulated within the same framework, we will hereinafter use $\boldsymbol{\theta}$ to represent the vector of all trainable parameters in the surrogate models, encompassing both the network parameters and the equation parameters.

Regarding I/BCs, noisy measurements at the boundaries or the beginning can be treated according to their level of confidence. If the I/BCs are known and deterministic, they should be incorporated into $\mathcal{R}$. If the I/BCs are stochastic, they should be included in $\mathcal{D}$. This approach ensures that the constraints are appropriately handled based on the nature of the available data.

2.4 Model Setup and Training Details. Several forward and inverse phonon transport problems are investigated to assess the feasibility and performance of the proposed method for solving phonon BTE with sparse, noisy data. To clearly illustrate the essence of our method while keeping the training efficient, we adopt the gray model [33], which assumes that phonons of all polarizations and frequencies are identical, sharing a constant group velocity. Despite its simplicity, the gray model has provided valuable insights into phonon transport behaviors [34]. Future research can extend this work to the nongray BTE. We examine several heat conduction problems for one-dimensional (1D) and two-dimensional (2D) geometries using gray model silicon (with parameters: group velocity $|\mathbf{v}|$ of 2677 m/s, relaxation time τ of 39.9 ps, and heat capacity C_V of 1,627,600 J/(Km^3)). Temperature or heat flux data derived from analytical or numerical solutions based on the gray BTE is used as synthetic data to simulate data from thermal measurements in practical setups. To demonstrate the effectiveness of the PI-BNNs in handling noisy data, the observed data are corrupted with Gaussian noise. In forward problems, the BTE solution is obtained even with noisy boundary conditions. In inverse problems, in addition to solving the BTE, we infer the unknown Knudsen number (Kn $= |\mathbf{v}|\tau/L$, L is the characteristic length and $|\mathbf{v}|\tau$ is the phonon mean free path) from noisy data.

Figure 1 illustrates the architecture of the PI-BNNs for solving the gray phonon BTE. In the gray model, the reduced energy distribution e depends on $\mathbf{x}$, $\mathbf{s}$, and t, which serve as inputs to the DNN. Each particle in the ensemble contains a single DNN, with outputs being $[T, e^{\text{neq}}]$. The equilibrium part e^{eq} can be easily derived from T. Since T is independent of the direction $\mathbf{s}$, $\mathbf{s}$ is set to a zero vector when evaluating T. Additionally, as phonon frequency and polarization are not considered in the gray model, the integrals in the equations only involve integrating over the solid angle space. The equation residuals and data mismatch are used to construct the likelihood, with network parameters (and Kn in inverse problems) learned via SVGD.

For the 1D problem, we use a fully connected neural network with 3 layers and 30 neurons per layer. For transient and 2D problems, the network consists of 5 layers, each with 30 neurons. The Swish activation function [35] is applied in each layer except for the last one, where a linear activation function is used. In the SVGD algorithm, we adopt 5 particles trained for 4000 epochs using the Adam optimizer with an initial learning rate of 1×10^{-3}. The prior for the network parameters follows a generalized Student's t-distribution $lst\left(\mu, \sigma^2, \nu\right)$, where $\mu = 0$, $\sigma^2 = 0.05$, and $\nu = 2$. For inverse problems, the prior distribution of $\log(\text{Kn})$ is a Gaussian distribution $\mathcal{N}(0, 1)$. The data variance is treated as a random

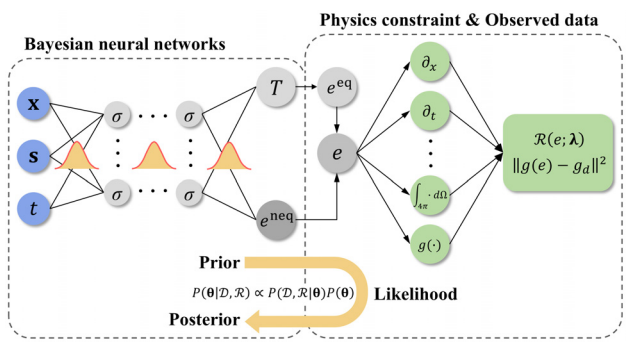

Fig. 1 Schematic of the PI-BNN scheme for solving forward and inverse thermal transport problems governed by the gray phonon BTE. Each particle employs a DNN to approximate the lattice temperature T and the nonequilibrium part e^{neq} of the phonon energy e. The inputs include spatial vector x, directional unit vector s, and time t. The equation parameter λ represents the Knudsen number Kn in this study. g is an observable function of e, and g_d represents the corresponding measurement data. σ denotes the Swish activation function. $p(\theta)$ is the prior for the DNN parameters and the unknown Kn in inverse problems, and $p(\mathcal{D}, \mathcal{R}|\theta)$ represents the likelihood of the predicted e. The posterior $p(\theta|\mathcal{D}, \mathcal{R})$ is derived based on Bayes' theorem. Initial and boundary conditions are imposed on either the equation residual $\mathcal{R}$ or the data mismatch $\mathcal{D}$, depending on the noise level.

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variable, with $\beta = 1/\sigma_{\mathcal{D}}^2$ following a conjugate prior $\text{Gamma}(\beta\,|a_0, b_0)$ with $a_0 = 2$ and $b_0 = 1 \times 10^{-4}$. The equation residual is assumed to follow a Gaussian distribution with variance $\sigma_{\mathcal{R}}^2 = 1 \times 10^{-4}$. For the training collocation points in 2D geometries, we sample the interior domain using quasi-random low-discrepancy Sobol sequences [36]. To compute the integrals, we use Gauss-Legendre quadrature [37] for the solid angle, with the number of abscissas set to 16 for the angular variable in 1D cases and 256 in 2D cases. The reference temperature T_{ref} in all test cases is set to 300 K. All neural networks are trained using PyTorch [38] on an NVIDIA RTX A6000 graphics processing unit (GPU).

3 Numerical Results

3.1 Forward Problems. In this section, we evaluate the proposed algorithm through two forward steady-state benchmark problems with limited noisy data. The first case involves a 1D cross-plane phonon transport problem (Fig. 2(*a*)) with noisy boundary conditions. To further validate the model performance, we then examine a 2D in-plane phonon transport problem (Fig. 2(*b*)) with noisy boundary conditions. Both problems are investigated using synthetic data generated from analytical solutions, which are intentionally corrupted at two different noise levels to assess the algorithm's robustness and accuracy.

3.1.1 One-Dimensional Cross-Plane Phonon Transport. We first evaluate steady-state thermal transport perpendicular to a silicon thin film (cross-plane direction) with a thickness comparable to the phonon mean free path (Fig. 2(*a*)). Heat conduction in this scenario is described by the 1D phonon BTE with thermalizing boundary conditions at both ends (detailed in Appendix). The film thickness is denoted as L, with temperatures $T_h = T_{\text{ref}} + \Delta T/2$ and $T_c = T_{\text{ref}} - \Delta T/2$ set at the left and right boundaries, respectively. The temperature difference ΔT is set to 2 K. The remaining boundaries are set to be periodic. Given that the length and width of the thin film are much greater than its thickness, the cross-plane heat conduction reduces to a 1D problem. For training the PI-BNNs, the spatial domain (x-axis) is equally discretized into 40 training points.

In this forward problem, we aim to predict phonon transport with noisy temperature data, which includes only six temperature data points in the interior domain and the boundary temperatures (i.e., T_h and T_c). The boundary conditions are considered noisy to simulate practical scenarios where there are no true phonon blackbody walls, and interface effects between materials may be non-negligible. Temperature data are "observed" at locations indicated by crosses in Fig. 2(*a*). The synthetic dataset of interior measurements is generated by adding Gaussian noise to the noise-free data from the analytical solution, with noise similarly added to the boundary temperatures. Two noise levels are considered: $5\%\Delta T$ and $10\%\Delta T$.

Temperature and heat flux are derived from the mean of the predicted phonon energy, and the uncertainty of the predicted temperature profile is evaluated as the standard deviation from

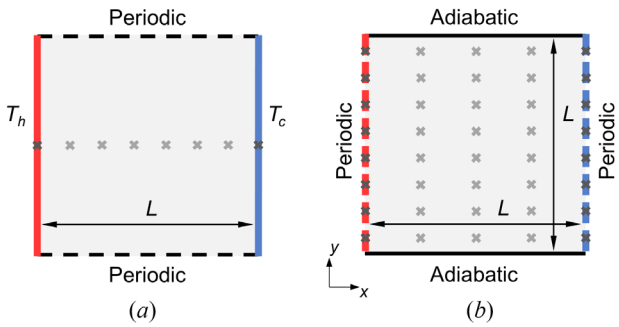

Fig. 2 Schematic of (*a*) cross-plane (1D) phonon transport and (*b*) in-plane (2D) phonon transport. Gray crosses represent internal data point locations, while black crosses denote boundary data points.

Eq. (15). Figure 3 shows the dimensionless temperature profiles at three different Kn values, where $T^* = (T - T_c)/\Delta T$ and $x^* = x/L$. Temperature profiles predicted by the PI-BNNs match well with analytical solutions [39], with absolute relative errors of less than 1% across both noise levels. The observed discrepancies generally fall within the standard deviation, which increases with higher noise levels. Notably, the temperature slips near the boundaries are attributed to size effects and nonequilibrium phonon transport. The predicted temperature slips can be influenced by noise in the boundary temperatures. However, the inclusion of interior data mitigates the impact of noisy boundary conditions, allowing for accurate forward simulations, whereas traditional deterministic solvers might be significantly compromised by such uncertainties.

Figure 4 shows the normalized cross-plane thermal conductivity $\kappa/\kappa_{\text{bulk}}$, where $\kappa = q \cdot L/\Delta T$ and $\kappa_{\text{bulk}} = C_{\text{V}}|\mathbf{v}|^2\tau/3$ is the thermal conductivity of bulk silicon. To verify the consistency of the proposed scheme and assess its sensitivity to the distribution of sparse and noisy data, we train five independent PI-BNNs (each with five particles) for each Kn. Different datasets, each corrupted with Gaussian noise at the specified level, are used for training. The mean thermal conductivity results from these runs are presented. In Fig. 4, we compare our results with both analytical solutions and experimental data [40] for silicon thin films at 300 K (where Kn is calculated using the model mean free path). It is observed that PI-BNN predictions closely align with the analytical solutions over a range of Kn values. While there is some deviation from experimental measurements in the diffusive regime (i.e., at low Kn), this can be attributed to the use of the gray phonon model, which simplifies the complexity of phonon transport in calculations. Nonetheless, the overall agreement remains strong. Error bars are not displayed as the standard deviations for all cases are below 3%. Additionally, the accuracy of heat flux prediction is not significantly impacted by the noise level, demonstrating the robustness of the PI-BNN in handling noisy data.

3.1.2 Two-Dimensional In-Plane Phonon Transport. The next case involves in-plane thermal transport. The schematic of the simulated object is illustrated in Fig. 2(*b*). A thin film with a width of L is bounded by two diffusely reflecting walls representing adiabatic upper and lower boundaries. Periodic boundary conditions are applied to the left and right boundaries, inducing a temperature gradient along the x-direction (in-plane direction). This problem is characterized by the 2D phonon BTE in the steady-state. For this scenario, we assume a uniform grid of 3×8 data points for the in-plane directional heat flux within the interior domain (Fig. 2(*b*)). There are also 2×8 equally spaced temperature data points at the left and right boundaries, separated by a distance of L, resulting in 8 data points representing the boundary temperature difference ΔT (with a true value of 2 K). All data are corrupted with Gaussian noise, with two noise levels considered: 5% and 10% of the data. For training, we use 500 Sobol points as the interior training points within a square-shaped region and 30 equally spaced points for the top and bottom boundaries, respectively.

Figure 5 shows the dimensionless x-directional heat flux $q_x^* = q_x/q_{\text{bulk}}$ at three different Kn values, where $q_{\text{bulk}} = \kappa_{\text{bulk}} \cdot \Delta T/L$. Although the difference increases with the noise level, the prediction results are consistent with analytical solutions [39]. Larger prediction errors and estimation uncertainties are observed for lower Kn values, attributed to the larger heat flux values for larger system sizes and the resulting greater data noise. The normalized thermal conductivity $\kappa/\kappa_{\text{bulk}}$ (with $\kappa = -\int_0^L q_x dy \cdot L/\Delta T$) is also evaluated. Figure 6 presents the mean results at different Kn values. For each Kn, the thermal conductivity is derived from 5 independent PI-BNNs trained on different datasets with random noise at the specified level. Error bars are not shown as all standard deviations are below 2%. The predicted thermal conductivity shows excellent agreement with analytical values from diffusive to ballistic regimes.

3.2 Inverse Problems. In this section, we test the proposed algorithm on three inverse problems with unknown Knudsen

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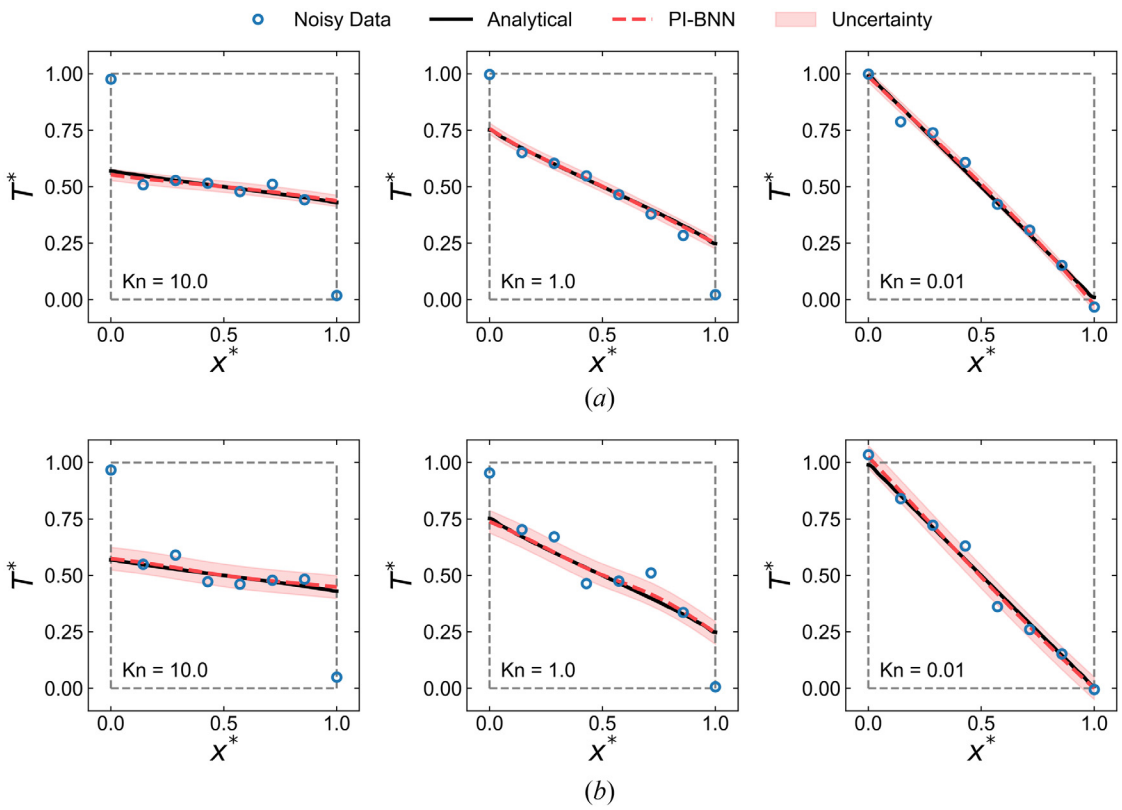

Fig. 3 One-dimensional cross-plane phonon transport–forward problem: dimensionless temperature profiles ($T^* = (T - T_c)/\Delta T$, $x^* = x/L$) at three Knudsen numbers with two data noise levels: (*a*) 5%ΔT and (*b*) 10%ΔT. Gray lines delineate the region $[0, 1] \times [0, 1]$ to better illustrate the temperature profiles and noisy boundary temperatures.

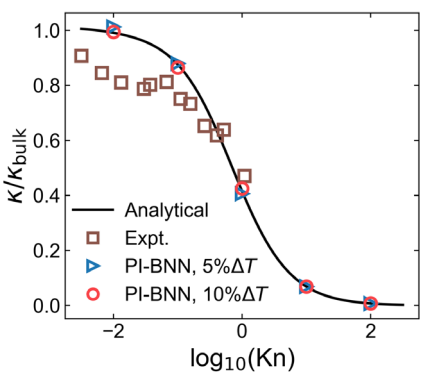

Fig. 4 One-dimensional cross-plane phonon transport–forward problem: normalized thermal conductivity κ/κ_{bulk} at different Knudsen numbers with two data noise levels. PI-BNN results are the mean of five independent runs trained on different datasets. Error bars are not shown as all standard deviations are below 3%. Experimental data [40] for silicon thin films at 300 K are also included for comparison.

numbers. Both steady-state and transient problems are investigated, given sparse and noisy data within the interior spatiotemporal domain. Our goal is to solve the phonon BTE while simultaneously identifying the unknown Kn, which is implicitly included in the BTE (Eq. (2)). We assume that the I/BCs are deterministic and noise-free (i.e., included in $\mathcal{R}$) for the inverse problems.

3.2.1 Two-Dimensional In-Plane Phonon Transport With Unknown Kn. This problem is similar to the previous example, except that Kn is treated as an unknown parameter, and the periodic boundary conditions are noise-free. Consequently, Kn is considered a random variable with a prior distribution. Given the scale of Kn, we learn $\log(\mathrm{Kn})$ and use a standard normal distribution as the prior. For the interior domain, we assume a uniform grid of 3×8 data points for the in-plane directional heat flux (Fig. 2(*b*)). For training, we use 500 Sobol points as interior training points, and 30 evenly spaced points for each of the boundaries.

Figure 7 illustrates the heat flux results at two noise levels. Although Kn is assumed unknown for PI-BNNs, the model predictions closely match the analytical solutions at different Kn values. Similar to the forward case, larger prediction errors and estimation uncertainties are observed for lower Kn due to greater data noise. Figure 8 shows the normalized thermal conductivity along with the inferred Kn by PI-BNNs. To evaluate consistency, each prediction is estimated as the mean result from 5 independent models using different datasets. As shown in Fig. 8, PI-BNN results agree well with analytical solutions in terms of both thermal conductivity and Kn. Most standard deviations from these independent runs are less than 3%, except for the case of $\mathrm{Kn} = 100$. The larger prediction error at $\mathrm{Kn} = 100$ can be attributed to the nature of the phonon BTE. In the context of the gray phonon model, the right-hand side of Eq. (2) becomes $-e^{\mathrm{neq}}/\mathrm{Kn}$, which is insensitive to Kn when it is large (e.g., over 100). As a result, greater prediction error and uncertainty in Kn are observed for cases at higher Kn, suggesting that the training is dependent on the dataset in this case. The predicted values of $\log_{10}(\mathrm{Kn})$ are displayed in Table 1. This table shows the maximum error among the five independent cases at different Kn values, which peaks at $\mathrm{Kn} = 100$. This suggests that the inferred Kn can be sensitive to the distribution of noisy data in the ballistic regime, especially when the dataset size is limited. Nonetheless, our predicted means still match well with the exact solutions, demonstrating the effectiveness of the PI-BNN in identifying the unknown equation parameter and quantifying the uncertainty arising from scattered noisy data.

3.2.2 Two-Dimensional Internal Heat Generation With Unknown Kn. In this scenario involving internal heat generation [41] (Fig. 9(*a*)), thermalizing boundary conditions with a

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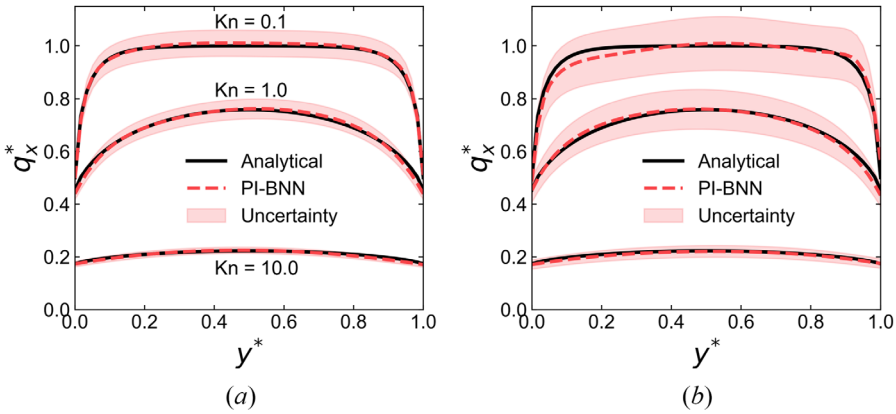

Fig. 5 Two-dimensional in-plane phonon transport–forward problem: *x*-directional heat flux ($q_x^* = q_x/q_{\text{bulk}}$, $y^* = y/L$) at three Knudsen numbers with two data noise levels: (*a*) 5% and (*b*) 10%

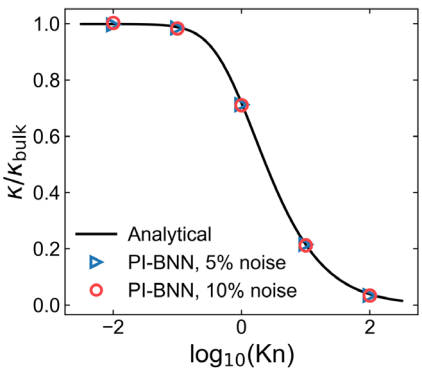

Fig. 6 Two-dimensional in-plane phonon transport–forward problem: normalized thermal conductivity $\kappa/\kappa_{\text{bulk}}$ at different Knudsen numbers with two data noise levels. Each PI-BNN result is the mean of five independent runs trained on different datasets. Error bars are not shown as all standard deviations are below 2%.

constant temperature $T_c = T_{\text{ref}}$ are set at the left and right boundaries, while the top and bottom boundaries are specularly reflecting. A uniform heat generation is applied across the entire simulation cell. We assume that the exact Kn is unknown, and we have a uniform grid of 3×8 data points for temperature, as marked by crosses in Fig. 9(*a*). We study two noise levels: 5% and 10%. For training, we use 500 Sobol points as interior collocation points in a square-shaped domain and 30 evenly spaced points for each boundary.

Figures 9(*b*) and 9(*c*) shows the dimensionless temperature T^* of the internal heat generation problem, defined as $T^* = 8\kappa_{\text{bulk}}(T - T_c)/(4\pi\dot{q}L^2)$. PI-BNN predictions agree well with numerical results [41] at different Kn values. The difference between the posterior estimation and the numerical solution is mostly bounded by the standard deviation. Larger errors and uncertainties in temperature for higher Kn are due to the larger magnitude of data noise, determined by the scale of temperature profiles. Notably, even with 10% noise and unknown Kn, our scheme provides reasonably accurate solutions. Table 2 shows the inferred Kn from 5 independent models trained on different datasets. Prediction error and variance increase with the noise level. PI-BNN performs well with low data noise at low Kn, but the impact of the data is more significant at high Kn due to the greater noise scales. Overall, the predicted Kn values match the true values.

3.2.3 One-Dimensional Transient Thermal Grating With Unknown Kn. Finally, the PI-BNN is verified with the 1D transient thermal grating (TTG) [42] problem with unknown Kn (depicted in Fig. 10(*a*)). We consider the thermal relaxation process of a 1D TTG, where a spatially sinusoidal temperature profile is initially set: $T(x, t = 0) = T_{\text{ref}} + A_0 \cos(2\pi x/L)$. A_0 is the initial amplitude of the temperature variation, and L is the grating period. Initially, all phonons are in equilibrium at the temperature of $T(x, 0)$. As time progresses, the amplitude of the sinusoidal temperature decreases. Given a weak laser pulse in TTG, the temperature can be

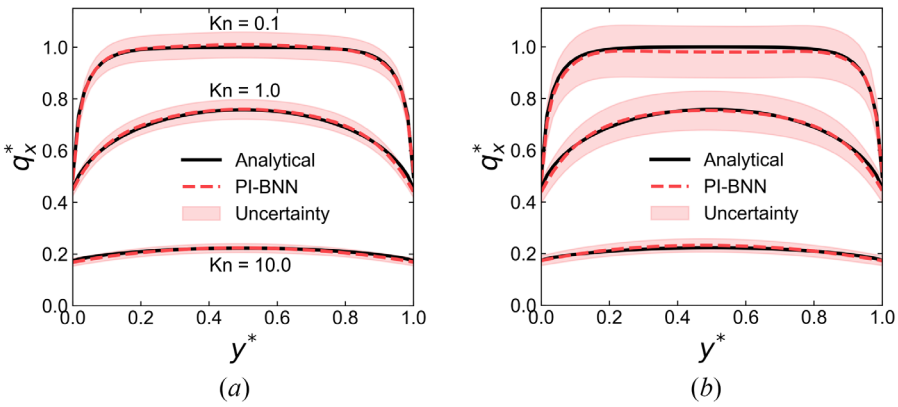

Fig. 7 Two-dimensional in-plane phonon transport–inverse problem: *x*-directional heat flux ($q_x^* = q_x/q_{\text{bulk}}$, $y^* = y/L$) at three Knudsen numbers with two data noise levels: (*a*) 5% and (*b*) 10%. The Knudsen numbers shown here are the true values, which are assumed unknown and to be inferred by PI-BNNs.

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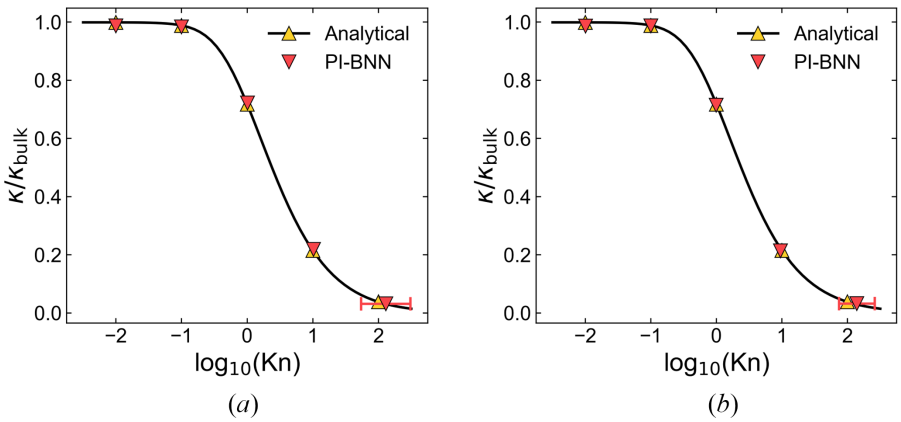

Fig. 8 Two-dimensional in-plane phonon transport–inverse problem: normalized thermal conductivity $\kappa/\kappa_{\text{bulk}}$ at different Knudsen numbers with two data noise levels: (*a*) 5% and (*b*) 10%. For PI-BNNs, Knudsen numbers are inferred from training. Each PI-BNN result (including thermal conductivity and Knudsen number) is the mean of 5 independent runs trained on different datasets. Most error bars are not shown as the standard deviations are below 3%, except for the one at Kn=100.

Table 1 Two-dimensional in-plane phonon transport: Predicted mean and standard deviation of $\log_{10}(\text{Kn})$ from sparse, noisy data using PI-BNNs

Level of noise	True $\log_{10}(\text{Kn})$	Mean $\log_{10}(\text{Kn})_{\text{pred}}$	Std	Max absolute error
5%	−2.0	−2.002	0.005	0.009
	−1.0	−1.003	0.005	0.011
	0.0	0.004	0.007	0.015
	1.0	1.010	0.022	0.039
	2.0	2.111	0.375	0.518
10%	−2.0	−1.998	0.004	0.007
	−1.0	−1.006	0.004	0.006
	0.0	−0.002	0.009	0.015
	1.0	0.977	0.028	0.056
	2.0	2.142	0.272	0.496

For each Kn, the result is evaluated from five independent models trained on different datasets. Maximum absolute error represents the greatest deviation from the true value among these independent cases.

Table 2 Two-dimensional internal heat generation: Predicted mean and standard deviation of Kn from sparse, noisy data using PI-BNNs

Level of noise	True Kn	Mean Kn_{pred}	Std	Max absolute error
5%	0.05	0.051	0.001	0.002
	0.1	0.099	0.001	0.002
	0.5	0.504	0.003	0.008
	1.0	0.977	0.024	0.041
	2.0	1.858	0.190	0.312
10%	0.05	0.051	0.001	0.002
	0.1	0.098	0.002	0.003
	0.5	0.491	0.012	0.023
	1.0	1.053	0.039	0.083
	2.0	1.712	0.258	0.619

For each Kn, the result is evaluated from 5 independent models trained on different datasets. Maximum absolute error represents the greatest deviation from the true value among these independent cases.

approximated as $T(x,t) = T_{\text{ref}} + A(t)\cos(2\pi x/L)$. We study one grating period and set the left and right boundaries as periodic. For the training data, we assume 8 equally spaced sensors on the x-axis to collect temperature at 10 moments (with equal intervals) over the time range of interest, resulting in 8×10 temperature data points. Two noise levels are considered: 5% and 10% Gaussian noise. As for training, a uniform grid of 1×100 is used for space discretization, and 60 time points with equal intervals are adopted.

Figures 10(*b*) and 10(*c*) shows the amplitude of the temperature variation $\hat{A} = A/A_0$ at different times. ξ is defined as $\xi = 2\pi\text{Kn}$. t^* represents the dimensionless time, defined as t/τ. PI-BNN results are in excellent agreement with the analytical solutions [42] for

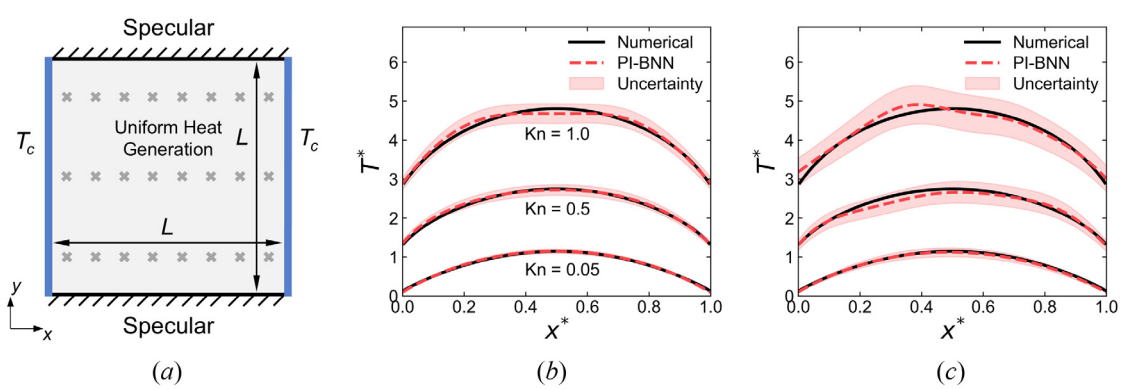

Fig. 9 Two-dimensional internal heat generation–inverse problem: (*a*) schematic with crosses indicating the locations of data points; dimensionless temperature profiles at three Knudsen numbers with two data noise levels: (*b*) 5%, and (*c*) 10%

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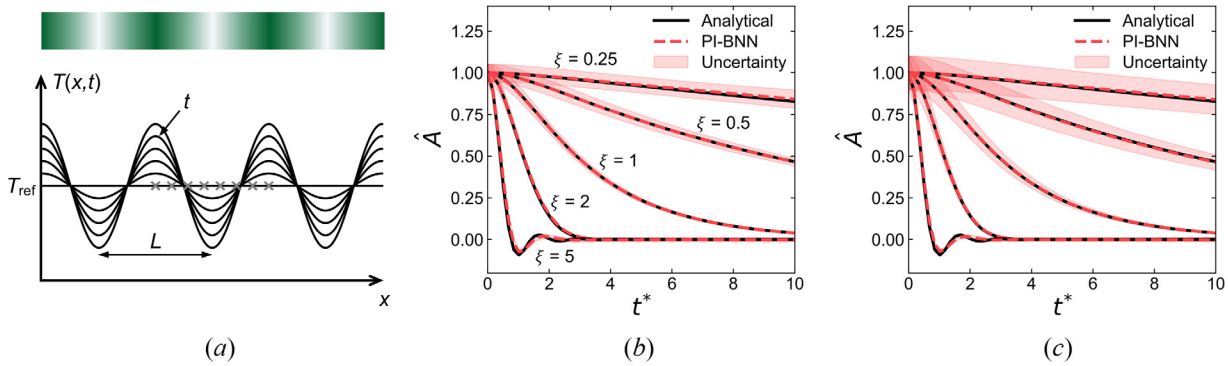

Fig. 10 One-dimensional transient thermal grating–inverse problem: (*a*) schematic with crosses indicating the spatial locations of data points; dimensionless amplitude of the temperature variation ($\hat{A}=A/A_0$, $t^*=t/\tau$) at different Knudsen numbers $\xi=2\pi$Kn with two data noise levels: (*b*) 5% and (*c*) 10%

Table 3 One-dimensional transient thermal grating: Predicted mean and standard deviation of ξ from sparse, noisy data using PI-BNNs ($\xi=2\pi$Kn)

Level of noise	True ξ	Mean ξ_{pred}	Std	Max absolute error
5%	0.25	0.248	0.005	0.011
	0.5	0.498	0.003	0.005
	1.0	0.998	0.003	0.006
	2.0	2.034	0.046	0.082
	5.0	5.135	0.112	0.287
10%	0.25	0.224	0.026	0.057
	0.5	0.498	0.006	0.011
	1.0	0.982	0.024	0.064
	2.0	1.943	0.063	0.145
	5.0	4.802	0.239	0.636

For each ξ, the result is evaluated from five independent models trained on different datasets. Maximum absolute error represents the greatest deviation from the true value among these independent cases.

different Kn values. The quantified uncertainty decreases with the scale of $\hat{A}$. The misalignment within the time range $t^* \in [1, 3]$ at $\xi = 5$ is likely due to the lack of training data in this region, as the temperature profile is sampled at only 10 moments evenly distributed across the time range for all cases. Table 3 shows the predicted ξ from 5 independent models trained on different datasets. PI-BNN works reasonably well at low Kn, but the sparsity of data at critical time points reduces accuracy at higher Kn. Nevertheless, the predicted Kn values are close to the true values across both noise levels.

4 Discussion and Conclusion

The performance of the proposed scheme can be influenced by the available dataset, particularly the noise level and the specific data distribution. Our observations indicate that higher noise levels lead to increased prediction error and estimation uncertainty. Additionally, the sparsity of the data can result in sensitivity to the specific dataset, causing inconsistencies when different datasets are used. For instance, in the inverse 2D in-plane heat conduction problem, larger error and variance in the inferred Kn were noted at high Kn, even with low data noise. The sampling locations may also affect the data distribution and subsequently the prediction accuracy. In the TTG case, without any assumption on the temperature decay curve, the lack of sampling around critical time points led to inaccuracy in capturing the trend of temperature changes at high Kn.

Efforts have been made to solve forward and inverse heat conduction problems based on Fourier's law (i.e., in the diffusive regime) using PINN-based models [43,44]. However, our earlier works have only investigated forward problems for nondiffusive heat conduction. Unlike macroscopic heat conduction problems, where temperature data are readily available, leveraging micro/nanoscale thermal measurements for simulating nondiffusive thermal transport requires further consideration. To seamlessly integrate the current framework with thermal measurements and enable realistic thermal simulation and inverse modeling, several improvements are needed for enhanced performance, which warrants further studies. First, the current scheme should be extended to solve mode-resolved phonon BTE where the gray model is not sufficient. While it is straightforward to include additional phonon properties as input to the network, temperature observations alone may not suffice to uniquely determine the mode-specific solution, which requires further investigation. Second, using more realistic measurement data and considering additional sources of noise, such as physical model inadequacy or source term noise, even noise in data spatiotemporal locations, is crucial to validate and improve the proposed scheme for practical applications. Third, evaluating the performance of PI-BNNs in more complex geometries (e.g., 3D transistor structures) and in inverse modeling for system parameters other than Knudsen number or characteristic length would be interesting.

In summary, we propose a physics-informed Bayesian deep learning framework for solving phonon BTE-governed forward and inverse heat conduction problems with sparse and noisy data. This framework integrates a Bayesian neural network for the solution, a likelihood function based on measurement uncertainty and equation constraints, and the SVGD algorithm for learning posterior distributions and quantifying uncertainty. Despite limited and noisy data, the proposed method is capable of reconstructing temperature or heat flux profiles and inferring unknown Knudsen numbers. Numerical experiments conducted on various forward and inverse problems demonstrate its effectiveness in modeling phonon transport across ballistic to diffusive regimes. By examining the impact of noise levels and data distribution, we have gained deeper insights into the accuracy and robustness of our method. This work represents the first effort to integrate noisy data into simulations of nondiffusive thermal transport governed by the BTE, particularly for inverse modeling. Our findings could significantly impact thermal transport research, making the BTE a viable tool for inverse design and optimization.

Author Contribution Statement

R.L., J.Z., and T.L. conceived the idea and initiated this project. R. L. designed and tested the model. All authors have contributed to the writing of the paper and the analysis of the data.

Acknowledgment

The simulations are supported by the Notre Dame Center for Research Computing, and NSF through the eXtreme Science and

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Engineering Discovery Environment (XSEDE) computing resources provided by Texas Advanced Computing Center (TACC) Stampede II under grant number TG-CTS100078. We also thank Professor Jian-Xun Wang for providing GPU computing facilities for testing our model.

Funding Data

- Office of Naval Research (ONR) MURI (award ID: N00014-18-1-2429; Funder ID: 10.13039/100000006).
- Defense Advanced Research Projects Agency (DARPA) (award ID: HR00112390112; Funder ID: 10.13039/100000185).

Conflict of Interests

The authors declare no competing interests.

Data Availability Statement

All data needed to evaluate the conclusions in the paper are present in the paper. Additional data related to this paper may be requested from the authors.

Appendix: Boundary Conditions

Four kinds of boundary conditions are usually met in phonon transport problems.

(1) In dealing with the thermalizing boundary, we assume all phonons emitted by the boundary are in thermal equilibrium with the predefined boundary temperature T_b, which is formulated as

$$e(\mathbf{s}) = e^{\mathrm{eq}}(T_b), \quad \mathbf{s}\cdot\mathbf{n}_b > 0 \tag{A1}$$

where $\mathbf{n}_b$ is the normal unit vector pointing into the computation domain.

(2) The specularly reflecting boundary condition is an adiabatic boundary condition in which the reflection angle of the phonon is equal and opposite to the incident one, given by

$$e(\mathbf{s}) = e(\mathbf{s}'), \quad \mathbf{s}\cdot\mathbf{n}_b > 0 \tag{A2}$$

where $\mathbf{s}' = \mathbf{s} - 2(\mathbf{s}\cdot\mathbf{n}_b)\mathbf{n}_b$ is the incident direction. Therefore, the phonon energy is conserved and there is no energy transfer across the boundary.

(3) The diffusely reflecting boundary condition is another adiabatic boundary condition, where the net heat flux is zero. At such boundaries in the gray model, the phonons are reflected with equal probability along all possible directions, namely,

$$e(\mathbf{s}) = \frac{1}{\pi}\int_{\mathbf{s}'\cdot\mathbf{n}_b<0} e(\mathbf{s}')|\mathbf{s}'\cdot\mathbf{n}_b|d\mathbf{\Omega}, \quad \mathbf{s}\cdot\mathbf{n}_b > 0 \tag{A3}$$

(4) For the periodic boundary, a phonon that crosses it is emitted at the opposite boundary with the same velocity vector. Additionally, two corresponding boundaries follow the local thermal equilibrium,

$$e(\mathbf{x}_{b_1}) - e^{\mathrm{eq}}(T_{b_1}) = e(\mathbf{x}_{b_2}) - e^{\mathrm{eq}}(T_{b_2}) \tag{A4}$$

where $\mathbf{x}_{b_1}$, T_{b_1} and $\mathbf{x}_{b_2}$, T_{b_2} are the spatial coordinates and temperatures of two associated periodic boundaries b_1 and b_2, respectively.

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